B.1.2 Characteristics of better services: What were we missing?

The big idea that paved the way for an entire dimension of solutions was waiting to be discovered. It is often said that the UNIX philosophy is “do one thing and do it well,” and in some ways, microservices follow this idea of coherence in their design. But this is not unique to the microservices architecture; SOA had this mentality from the beginning. The key was the same idea, but in a different dimension: the infrastructure.

The fundamental trigger is containerization. Containers arise from the idea of jailing an operative system in a prison by dividing it into folders that become cells in which each user can act like a root. They are not root, but within the confines of their “prison,” they behave as such. The idea was first implemented in 1979 in version 7 of UNIX with the `chroot` command. It was integrated into BSD with some extensions in 1982 and matured into FreeBSD in 2000. Only then did it have a reliable degree of isolation.

In 2006, a company called Joyent created the infrastructure to provide containers to its users but received little attention, although it continued to innovate in this space. That same year, AWS introduced EC2. In 2013, dotCloud, which was working in this area and failing as a company, opened its code and partnered with Red Hat. The company’s new name was Docker, Inc. The industry responded positively and embraced containers as a whole, which changed everything. Containers became the “new black.”

The jail had room for the essentials that EAIs, ESBs, and different SOAs craved: infrastructure composition in two dimensions and